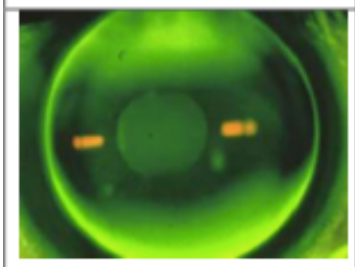
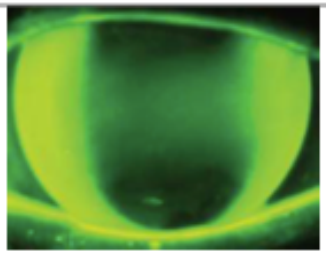
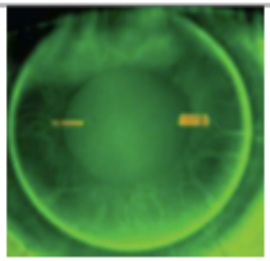
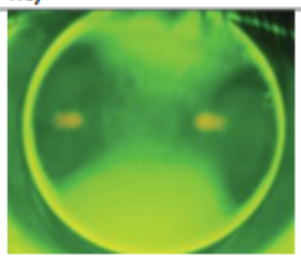
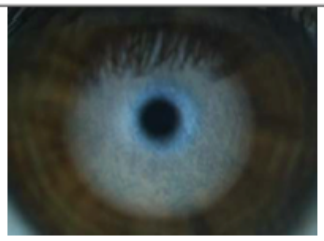
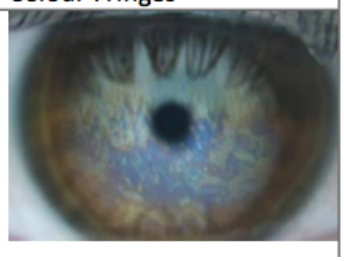
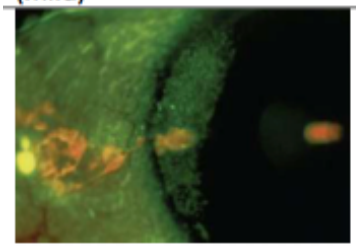
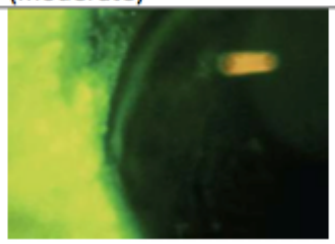
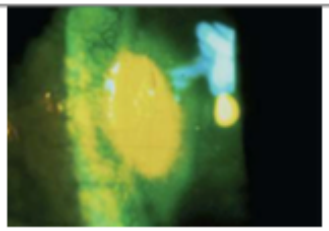
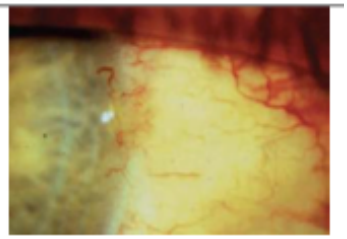
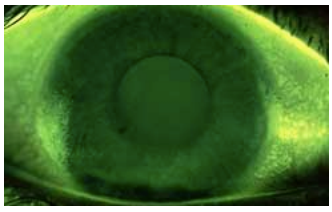
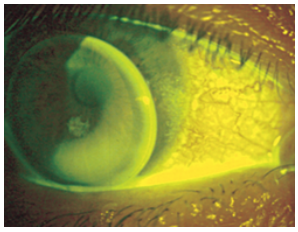
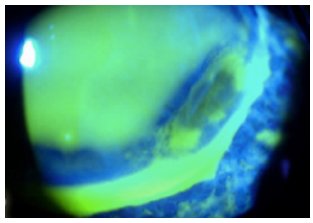
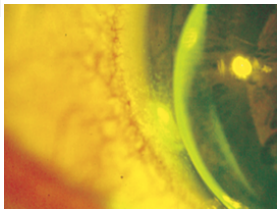
Fitting Characteristics of RGP

With the Rule	Against the Rule	Low WTR (Compromise fit)	High WTR (Compromise fit)
			

Tear Film Assessment

Open Meshwork	Closed meshwork	Wave	Colour Fringes
			

Physical Effects of CL

3 and 9 o'clock staining (mild)	3 and 9 o'clock staining (moderate)	Dellen	Vascularised Limbal Keratitis
			
			

3 and 9o'clock Staining

- Localised punctate corneal staining outside Rigid Lens at 3 and 9o'clock
- Due to desiccation (drying) of peripheral area of edge
    - Excessive Edge Lift- thirsty tear meniscus and sucks tears away
    - Inadequate edge lift- minimal tear reservoir
    - Bridging Effect at lens edge- eyelid cannot resurface cornea with mucus
- Symptoms are non-minimal even for MILD to SEVERE causes. May be a vague report of discomfort and dryness
- <u>Management</u> (From Clinical Cases Textbook)
    - Rule out solution preservative reaction by changing the lens care system
    - Refit RGP with smaller diameter
    - Increase wettability and change to higher Dk lens
    - Can consider Piggyback system with SCL and the same RGP
    - Punctal occlusion
    - Extensive lubrication with full time wear

Dellen

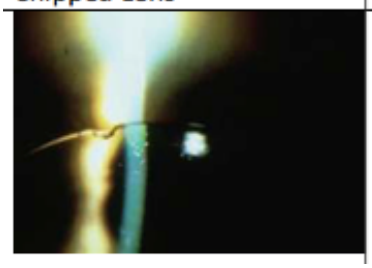
- Localised area of thinning or drying of the peripheral cornea
- Epithelium is compromised and staining has started in the stroma- dangerous

- Can rapidly become infected and lead to peripheral ulceration and scarring→ Need to take action
- Management
    - DDx out neutrophic ulcer and corneal infiltrate
    - Cease CL wear→ discontinue and refit with caution
    - Tx with eye lubrication, artificial tears and/or ointments 4x day and at bedtime
    - Could consider giving antibiotic prophylaxis: Ciprofloxacin 0.3% 4x day
    - Consider punctual plug
    - SCL bandage lens
    - Review in one week to confirm resolution

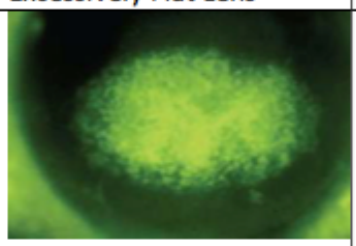
Vascularised Limbal Keratitis

- More chronic manifestation of 3 and 9 o'clock staining: weeks→ months
- Not as dangerous as dellen but takes a long time to recover
- Management
    - DDx out corneal ulcer, pterygium, phylectenulosis, viral and allergic conjunctivitis
    - Discontinue lens wear
    - Give Prophylaxis 4x day and follow up next day
    - Artificial lubricants 4x day
    - Refit RGPs with more peripheral lens clearance and central lens position (e.g. apical touch), higher Dk for better oxygenation
    - Use of rewetting drops
    - Monitor regression (ghost vessels) and progression (refilling)

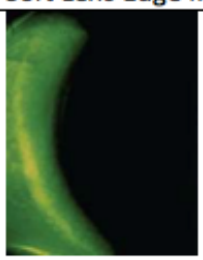
Chipped Lens



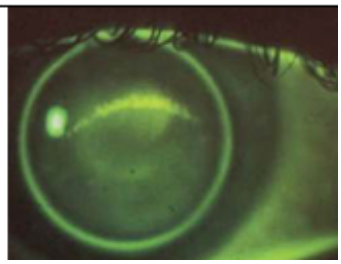
Excessively Flat Lens



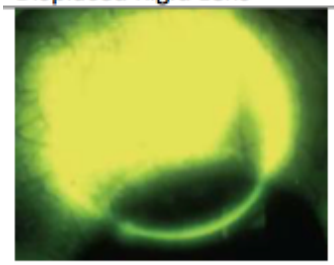
Soft Lens Edge Indent



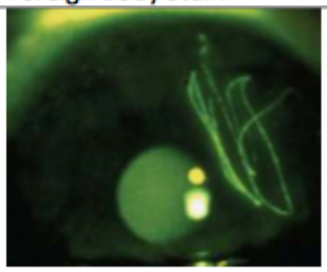
Lens Insertion Stain



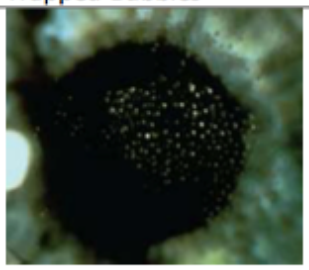
Displaced Rigid Lens



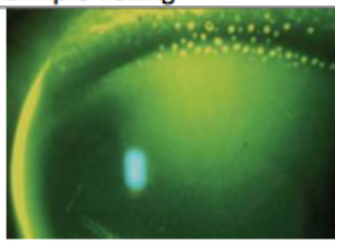
Foreign Body Stain



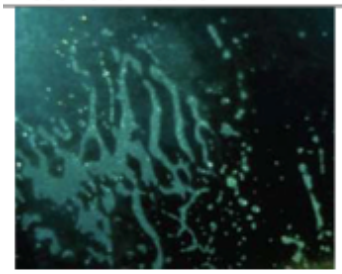
Trapped Bubbles



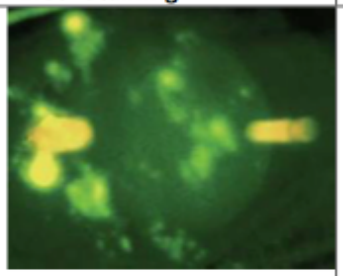
Dimple Veiling



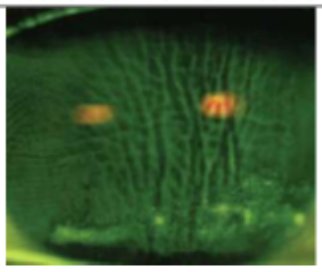
Post-Lens Debris



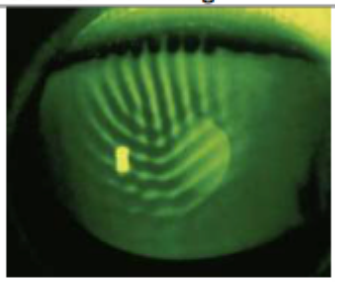
"Blot" Staining



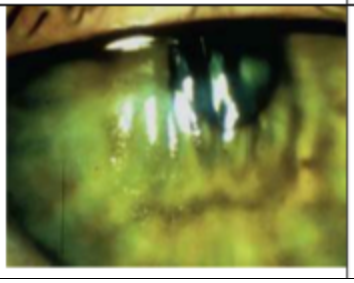
Fischer-Schweitzer Mosaic



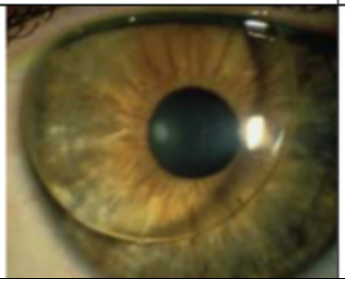
Corneal Wrinkling



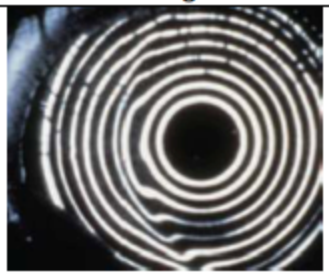
Wrinkled soft lens



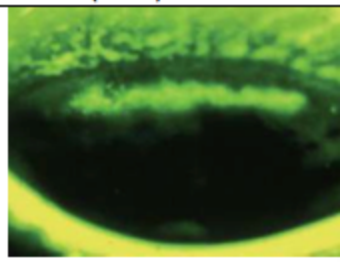
Adherent RGP lens



Corneal distortion from indentation ring



Superior Epithelial Arcuate Lesion (SEAL)



Chipped Lens

- Can give rise to traumatic epithelial staining

Excessively flat Lens

- May see this with excessively flat or mobile fit over time epithelium under centre of lens has been damaged
- Mechanical pressure on the cornea
- Signs: central epithelial edema and abrasion, Fl pattern often clarifies problem
- Management
    - Steepen BC
    - Increase OZD
    - Steepen peripheral curves

Soft Lens indent

- Typical lens edge indentation seen with an excessively tight soft lens, out on the conjunctiva in an arc where the edge of the lens has applied excessive pressure and rubbing

Corneal Staining Mechanical causes

- Lens handling
    - Rigid lens edge abrasion from insertion and removal
    - Fingernail abrasion
    - Suction cup lens removal in absence of lens
    - Soft lens removal in absence of lens: inferiorly and at 3 and 9o'clock
    - More likely in early Cl wear or experienced wear when they have gotten complacement
- Lens insertion stain→ need further teaching
- Displaced rigid lens→ need further teaching

Trapped Bubbles

- Caused by RGP lens that were fit too steep and dented the epithelium

Dimple Veiling

- Lens is still on the eye in the photo, if we remove it, the bright spots will still hold the dye but will disappear with saline flushing
- Signs and Symptoms:
    - Usually none
    - Slight blurring if many dimples in central location
- Management: Alter lens fit to avoid excessive inferior lens stand-off

Corneal Foreign Body

- Important to classify into metallic vs non-metallic, embedded vs superficial and penetrating vs non-penetrating
- Management
    - Instil NaFl and look for Seidel sign
    - Examine fornices/tarsus to rule out other FB
    - Instil fluorescein and irritate FB away with sterile saline
    - Record size, depth, location
    - Treat as an abrasion (i.e. topical antibiotics, NSAIDs, bandage CL, cycloplegia)
    - F/up in 24hrs, 3days, 5-7days, 12% RCE risk

Post-lens Debris

- Mucus and debris behind a SCL
- Abrasion is light and in a vague way on the surface, is not sharply defined
- Can also be called blot or dot staining

Corneal Indentation from adherent RGP lens

- Can clearly see the indentation ring where pressure has been applied
- Transient, within about 1hour it will be gone as the epithelium moves to fill in the trough
- While present, will cause a lot of distortion which may interfere with vision
- Disruption to the mucus layer where the lens had been sitting
- Can induce rapid peripheral desiccation→ dellen→ peripheral ulcers
- Due to thinning of post-lens tear film, increase tear viscosity
    - Eyelid pressure during closed eye lens wear→ flat-fitted lens, usually adhere nasally
    - Low-riding, relatively immobile lens, usually adhere inferiorly
- Signs and Symptoms
    - Immobile lens (rarely seen)
    - Indentation ring
    - Paracentral patch of dense punctate staining, peripheral arcuate desiccation staining

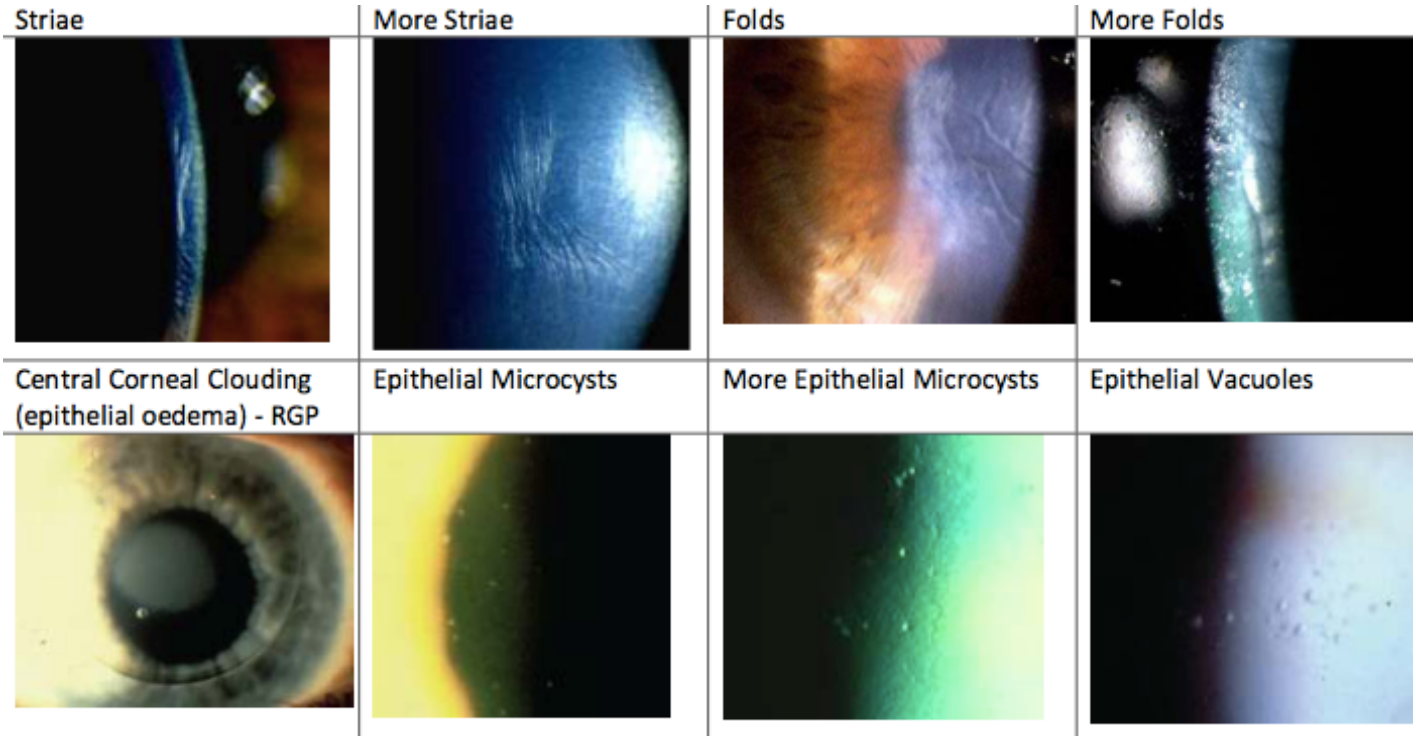
- Symptoms usually minimal→ dryness, slight blur. Can have sharp burning pain on regaining mobility (occasional)

- Management
  - Very frustrating
  - Effect of parameter variation unpredictable→ avoid flat lens fit
  - Change lens material
  - Counsel patient→ self monitoring and initiation of lens movement
  - Use of in eye lubricants
  - Modify blinking behavious
  - Avoid EW

SEAL

- Superior Epithelial Arcuate Defect
- Mechanical disturbances or trauma when wearing SCLs→ inadequate flexure in the superior corneal regions, combined with eyelid forces, found in tight or low positioned eyelids
- More common in SiH SCLs
- Can be asymptomatic especially if they are wearing the SCL which acts as a soft bandage lens
- Management
  - Cease CL wear until completely healed
  - Unpreserved ocular lubricant used every hour
  - Infection risk→ prophylaxis e.g. ciprofloxacin
  - Refit with lower modulus lens, switch lens design, flatter BC may help
  - RGPs may be a potential solution

Corneal Hypoxic Response / Corneal Edema



Striae

- Fine vertical or horizontal grey-white lines
- Resemble stromal nerves but do not bifurcate
- Located in posterior stroma
- Appear at >5% edema

Folds

- Dark lines at level of Decemet's Membrane
- Represent more severe buckling of posterior stroma
- Appear at >10% edema
- Suggests clinically unacceptable lebel of edema

Central Corneal Clouding

La Hood and Grant (1990):
- corneal edema estimated by counting striae and folds
- provides clinical means for estimating edema levels

- Rule-of-thumb:

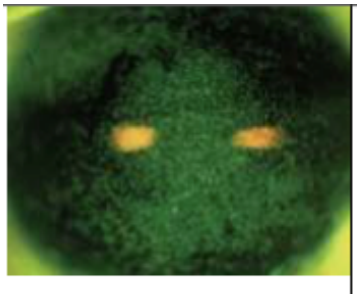
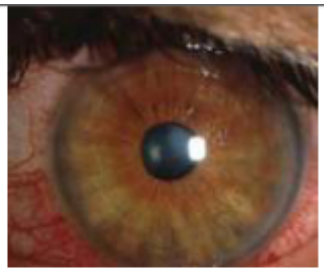
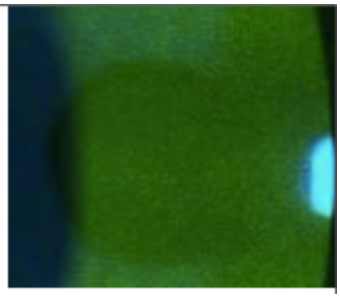
1 stria	=	5% edema
5 striae	=	8% edema
10 striae	=	11% edema
1 fold	=	8% edema
5 folds	=	11% edema
10 folds	=	14% edema

- From acute epithelial hypoxia caused by a change in epithelial metabolism creating lactate in the stroma which is pooling tears into the epithelium
- Epithelial edema giving rise to cloud of haze

Epithelial Microcysts

- Use high mag, marginal retro-illumination, reverse illumination→ on the side closest to the illumination appears dark and opposite side lights up. (Vacuoles unreversed and look like bubbles not dust)
- 15-50um in diameter, like specks of dust
- Distributed in annulus in mid-periphery, typical lication near the pupil margin
- Consists of pockets of disorganised or dead cellular material
- Clinical Significant
    - EW>> DW (rare0
    - Hydrogels >> SiH or RGP (depending on Dk/t)
    - Clinical intervention needed if more than 50 microcysts, or if they are breaking through the epithelium
- Management: Remove the px from the CL wear until the microcysts settle down and refit into another CL with less hypoxic stress

Adverse Response to Lens Care Products

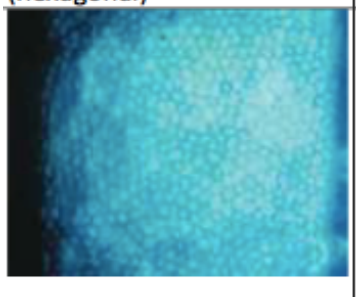




CL solution Toxicity/Sensitivity

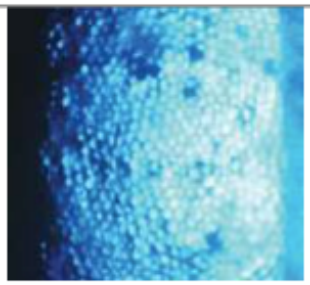
- More common in MPS than peroxide based solutions and more in SiH lenses
- Clinically manifests as discrete scattered punctate staining and all over cornea or in an annular pattern near the limbus
- Peak occurrence ~2hrs after CL insertion
- Associated with conjunctival hyperemia and edema
- Sx: asymptomatic to slight discomfort
- If acute and severe pain, tearing, likely to be non-neutralised peroxide in the eye (Tx as grade 1 chemical burn)
- Management: reteach, change care regiment→ preservative free?

Endothelial Response to CL

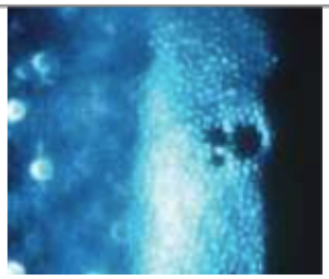
Normal Endothelium (hexagonal)



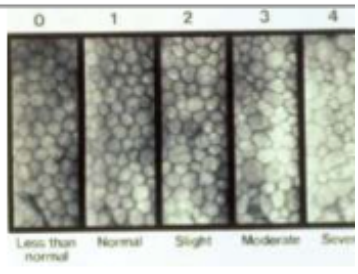
Endothelial Blebs



Endothelial Guttata

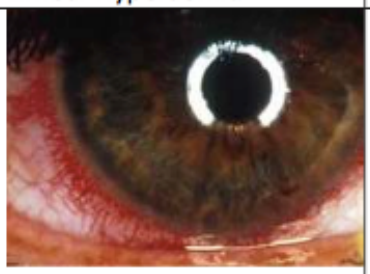
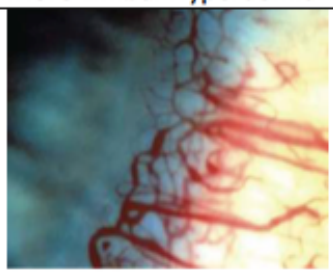
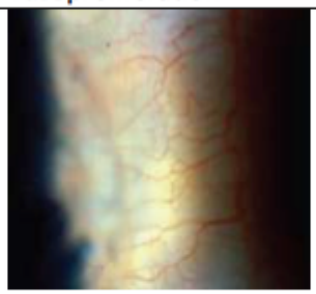
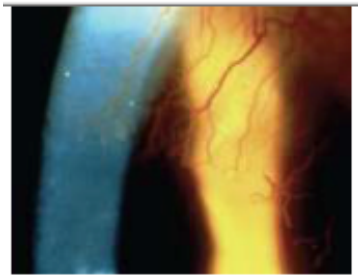
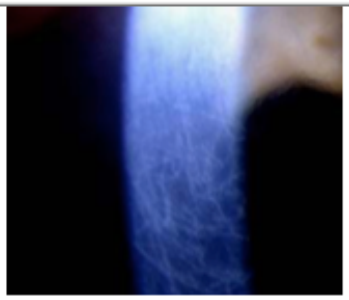
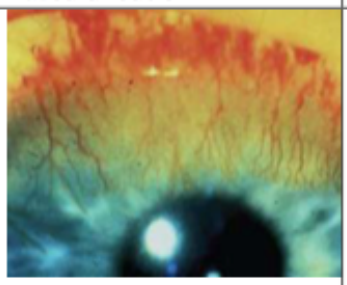
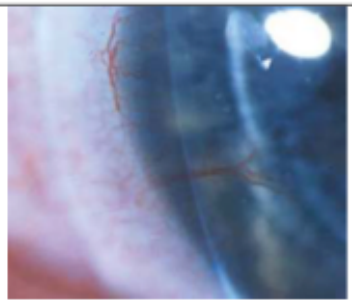
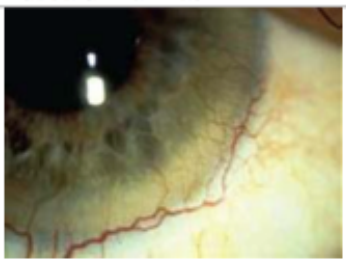


Endothelial Polymegithism



Clinical Mx of endothelial Changes

- Increase corneal oxygenation
    - Reduced hypoxia→ reduced lactate and minimise stromal acidosis
- Increase CO2 efflux from the eye
    - Increase lens Dk/t(CO2) by increasing Dk/t(O2)

Inflammatory Responses			
Conjunctival Hyperaemia	Limbal Hyperaemia	More Limbal Hyperaemia	Vasoproliferation
			
Neovascularisation	Ghost Vessels	Superficial Vascularisation	Stromal Vascularisation
			
Vascular Pannus			
			

Conjunctival Hyperemia

- dilation of conjunctival blood vessels→ increased blood flow→ increased 'redness'
- represents low-grade inflammatory response – acute (reactive) →chronic
- normal diurnal variation, daily lens wear → slight increase in evening
- overnight lens wear (extended/continuous)→ significantly worse immediately on awakening
- record using grading scales, specify quadrant

Limbal Hyperemia

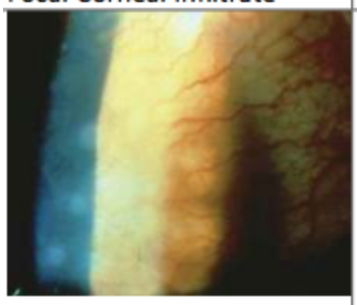
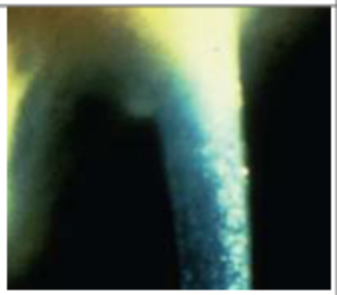
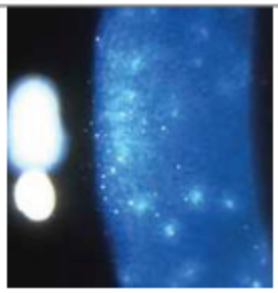
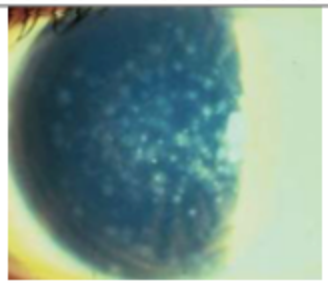
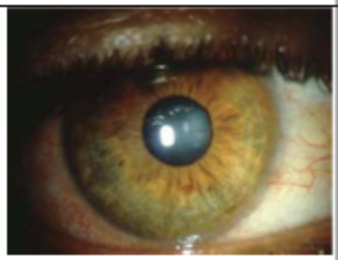
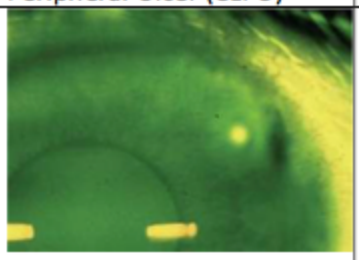
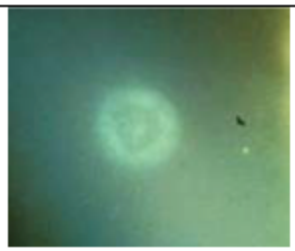
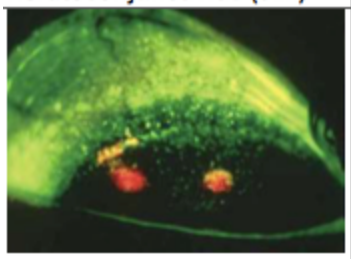
- Clinical manifestation:
  - dilation of pre-existing limbal blood vessels (not new blood vessel growth) → increased blood flow → increased 'redness'
  - may be acute or chronic
- Prevalence of response:
  - rigid lens wearers » non-lens wearers
  - hydrogel lenses >> SiH or rigid lenses
  - extended wear > daily wear
- Possible CL related etiology
  - Mechanical trauma
  - Physical restriction of limbal vessel drainage
  - Acute or chronic hypersensitivity response

Corneal Vascularisation

- Critical sign is development of spikes or buds projecting from normal limbal arcades
- Any vessel ingrowth into cornea beyond translucent limbal transition zone is cause for concern
- <u>Management</u>
  - Draw diagram→ location, extent, depth
  - Use grading scales CCLRU
  - Photograph or video for later reference

- Discontinue lens wear
- Monitor regression of BV (ghost vessels)
- Eliminate all possible predisposing factors
- Resume lens wear with extreme caution
  - Increase oxygenation
  - Optimise lens fit
  - Modify lens care regimen
- Close monitoring required (refilling of ghost vessels)

Infiltrative Responses

Focal Corneal Infiltrate	Diffuse Corneal Infiltrate	Severe Infiltrative Keratitis	Severe Infiltrative Keratitis
			
Contact Lens Acute Red Eye (CLARE)	**CLARE 24hrs Later**	**Active Contact Lens Peripheral Ulcer (CLPU)**	**Non-Active CLPU**
			
Superior Limbic Keratoconjunctivitis (SLK)	**More SLK**		
			

Corneal Infiltrates (Infiltrative Keratitis)

- Acute inflammatory corneal response
- Inflammatory cells from limbal vasculature invade the corneal tissue
- Cells are mainly polymorphonuclear (PMN) leukocytes (may include macrophages, lymphocytes)
- Rare in non-lens wearers and RGP wearers
- EW >> Daily wear
- 2nd stage of inflammatory cascade
- Clinical Presentation
  - Epithelial, sub-epithelial or stromal
  - Focal or diffuse (CL is usually focal)
  - 0.5-2mm in diameter round, grey-white
  - usually separated from the limbus by a clear zone
  - may be asymptomatic in a quiet eye
  - Can be associated with inflammatory sx e.g. limbal hyperemia in the same sector, pain, discomfort
  - If staining overlying cornea→ may be infection
- Etiology
  - Chemotactic stimulus for leukocyte invasion e.g. viral, toxic, allergic, obscure

    - CL related→ solution toxicity, poor lens fit, preservatives, toxic environmental stimuli, exo/endotoxins
- Management
    - If overlying NaFl staining→ assume corneal infection and refer for medical advice
    - Discontinue lens wear until infiltrates resolve (usually 1-2weeks to 3months)
    - Identify and eliminate predisposing factos
    - Resume wear with fresh lenses, increase disposal→ DD?
    - Monitor closely for recurrence
- Management for STERILE corneal infiltrates (Lily Ho)
    - Exclude infection
    - Remove CL immediately
    - Treatment is mostly palliative: Ocular lubricants/unit dose saline rinsing/cold compresses
    - If symptomatic and/or severe
        - Topical steroids e.g. Flarex 0.1% 4 x day
        - Prophylatic topical antibiotics
        - Review 24-48hours
        - Cycloplegia if AC reaction
        - Treat blepharitis if underlying causation

Contact Lens Acute Red Eye (CLARE)

- Acute inflammatory response of cornea immediately following eye closure
- Associated with severe conjunctival hyperemia
- Also called tight lens syndrome, non-ulcerative keratitis
- ONLY WITH SCL EXTENDED WEAR
- HWC>LWC, lens care, schedule
- 50% recurrence rate
- Clinical presentation
    - Sudden onset in middle of night→ DURING overnight lens wear. Awakened by symptoms or noticed soon after waking
    - Usually unilateral 88%
    - Symptoms: ocular pain (mild→mod), redness, tearing, photophobia
        - Symptoms rapidly subside once lens is removed
    - Signs: hyperemia, subepithelial infiltrates, rare AC response, minimal staining
    - Refer if overlying staining with NaFl
- Etiology
    - Gram –
    - Mostly H influenza
    - Carriers 5x more likely to develop CLARE
    - P. aeruginosa
    - S. Pneumoniae (gram +)
- Management
    - See px ASAP (same morning)
    - Lens removal→ sx relief, Cold compresses can help initially
        - Self-limiting if remove lens immediately
        - Cease CL wear
        - Careful monitoring over 24hours
    - Refer if significant staining or AC reaction
    - Therapeutics
        - Palliative: ocular lubricants non preserved
        - Consider mild steroid to hasten resolution
    - Monitor at intervals over next 48-72hours
        - Most symptoms resolve in days
    - Discontinue lens wear until any infiltrates resolve
        - Usually remain for 1 week
    - Identify and eliminate predisposing factors
        - Poor lens fit or condition
        - Compliance
        - EW, non-compliant with wear schedule
    - Recommend CL cautiously

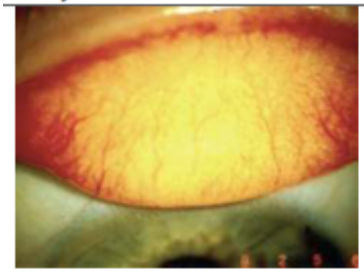
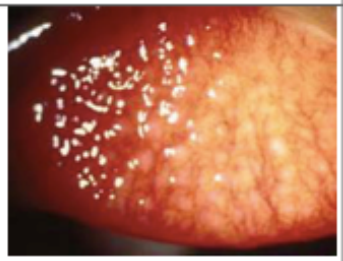
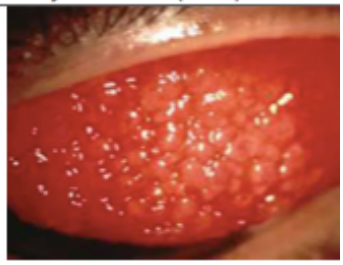
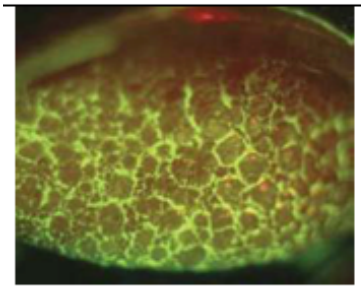
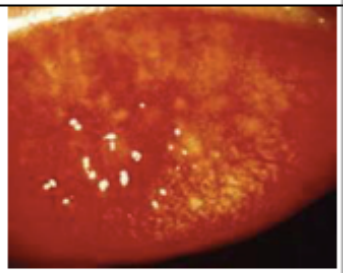
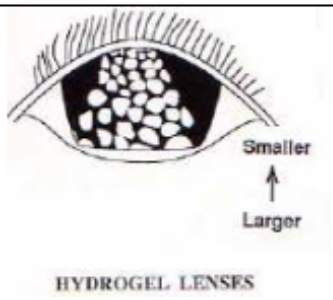
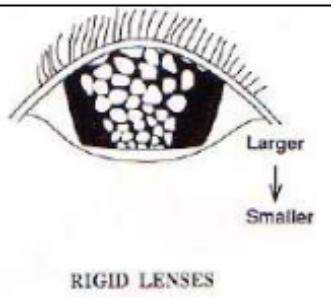
Contact Lens Peripheral Ulcer (CLPU)

- Acute inflammatory corneal reaction
- Presumed to be culture-neg (sterile)
- Mainly with SCL EW
- Relatively benign, few sequelae
- Clinical presentation
    - Small <2mm circular lesion
    - Located in the peripheral/mid cornea
    - Full thickness of epithelium usually→ intense stromal leakage→ NaFl staining
    - Stromal/subepithelial infiltrates
    - No discharge, no AC reaction
    - Crucial to DDx from infected ulcer
- Signs and Symptoms
    - Severe to moderate pain, FB sensation, irritation or asymptomatic (less pain than MK)
    - Limbal and bulbar redness
    - Focal excavation of the epithelium, infiltration and necrosis of the anterior stroma
    - Small, circular (commonly 0.5-1mm with 77^ <1mm). Slight diffuse infiltrate extend from the limbus to beyond the focal infiltrate
    - Usually single but can be multiple
    - No or minimal AC reaction
    - VA unaffected
- Possible etiological factors
    - Inflammation reaction to endo/exotoxins (more gram +ve)→ S. Aureus most common→ S. pneumoniae
    - Still need more evidence for etiology
- <u>Management</u>
    - Cease CL wear immediately: Lesion usually resolves without intervention 1-2 weeks
    - Monitor closely over first 24 hours
    - Ocular lubricants
    - Antibiotics >4xday if epithelium is compromised. Add steroid for more severe e.g. flarex→ weigh risk
    - IF IN DOUBT→ treat as MK until proved otherwise
    - Sequelae limited→ no effect on vision, resolve with a ciricular scar
    - Resumption of CL wear usually successful
    - Modify CL wear, care, regimen etc
    - Chance of recurrence

Superior Limbic Keratoconjunctivitis (SLK)

- Inflammatory response at superior limbus, involving adjacent conjunctiva and cornea
- Occurs in SCL wearers using CHEMICAL DISINFECTION and/or PRESERVED SALINE
- Usually bilateral, symmetrical
- Symptoms: redness, FB sensation, lens intolerance, decreased VA, photophobia, tearing
- Signs
    - Superior bulbar conjunctival and limbal injection
    - Superior micropannus
    - Thickening, irregularity of superior conjunctival epithelium
    - Superior corneal epithelial opacification
    - Represents delayed hypersensitivity reaction to preservative (usually thimerosal)
- <u>Management</u>
    - Discontinue CL wear
    - Wait for complete resolution
    - Avoid thimerosal in lens care system or move to peroxide based or DD

Allergic Cases

Normal Palpebral Conjunctiva	Papillary Conjunctivitis	Palpebral Conjunctival hyperaemia	Giant Papillary Conjunctivitis (GPC)
			
CLPC with Fluorescein	**Local CLPC**		
			

Papillary Conjunctivitis

- Chronic inflammation of superior palpebral conjunctiva
- Characterised by papillary hypertrophy= elevated lumps
- Associated signs: increased hyperemia, excessive mucus discharge, usually bilateral
- Follows progressive course
- Precipitating factors
    - Atopy: VKC, SLK etc.
    - Mechanical trauma: ocular prosthesis, suture ends, elevated corneal scars
    - Contact lens wear (has both atopic, chemical deposits and mechanical)

Giant Papillary Conjunctivitis (GPC)

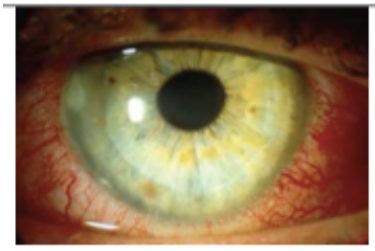
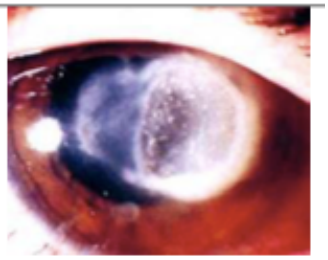
- Presence of giant >1mm papillae (normal is 0.1-0.2mm, 4-8papillae/mm)
- Can be end stage of papillary conjunctivitis

Contact Lens Papillary Conjunctivitis (CLPC)

- Associated with CL wear
- Represents early clinical manifestation
- If no intervention→ progression to GPC→ complete intolerance to CL wear
- Early detection and intervention essential for successful continuation of lens wear
- Clinical Symptoms
    - Follows a progressive course
    - Mild mucus discharge
    - Ocular itching, especially on lens removal
- Symptomatic stage
    - Increased stringy mucus discharge and watering
    - Increased lens movement, occasional displacement
    - Visual disturbance (mucus deposits, lens movement)
    - Intense itching and Cl awareness
- Clinical Signs
    - Increased palpebral conjunctival hyperemia due to increased blood vessel dilation (very leaky and edema)
    - Papillary hypertrophy in areas 1,2,3
    - Normally central and move concerning for CLPC
    - Staining on top of papillae as microvilli that are usually there will be damaged by the papillae
    - Mucus strands and sheets
    - Often associated with protein deposits on lens front surface (e.g. if lens is very dirty)
- Clinical detection
    - Signs to not equal symptoms
    - Careful history

- Lid eversion
- Clinical Management
  - Early detection
  - Appropriate intervention
  - Discontinue lens wear→ relief of Sx in 2-5days
  - News lenses of same material, design and increase frequency of disposal (DD)
  - Modify lens care and maintenance regimen→ increase compliance, enzymatic protein removal, avoid preserved solutions
  - Reduce lens wearing time (encourage to take CLs off when home)
  - Therapeutic intervention
    - Opticrom, Patanol, Zaditen= mast cell stabilisers (decrease histamine release, itching, decrease inflammatory signs and symptoms
  - Topical corticosteroids, NSAIDs
  - Tetracycline, vit A
- Prognosis
  - Total regression may take years
  - Recurrence with re-application of stimulus
  - Management aimed at continuing CL wear
- Local vs General CLPC
  - Rigid and SiH lens tend to induce localised CLPC→ suggest mechanical etiology (stiffer)
  - Hydrogel lenses more often associated with general CLPC
  - Atopic individuals more susceptible to general CLPC→ suggest immunological etiology
  - Both mechanical and immunological factors may play a role in some cases of CLPC

Corneal Infection

Rigid Lens Corneal Infection	Soft Lens Corneal Infection	Microbial Keratitis (Pseudomonas)	Acanthamoeba Keratitis
			

Microbial Keratitis

- Characteristics
  - Excavation of the corneal epithelium, Bowman's Layer and stroma
  - Infiltration and necrosis of tissue
    - Focal infiltrates usually large >1mm and irregular (but can start small)
    - Found in the central or paracentral region usually
  - Anterior chamber reaction
  - Severe limbal and bulbar redness
  - Rapid onset of moderate-severe pain
  - Decrease central VA
  - Tearing, mucopurulent OR purulent discharge
  - Photophobia
  - Puffiness of lids
  - Removing lenses does NOT provide relief reasonably rapidly
- Culture
  - ~50% culture proven
    - 56% P. Aeruginosa (most likely causative organism)
    - 5% Acanthamoeba
    - 5% yeast and fungi
    - 25% Gram +
    - 9% other environmental Gram –
  - Different if non-CL wearer

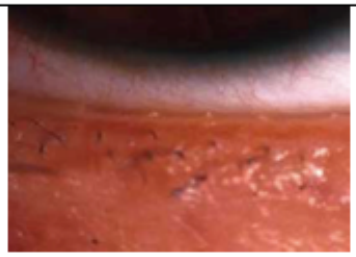
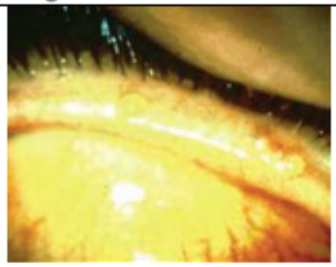
- Disease severity and outcome depend on microbe
- Environmental/climate factor

- Management
  - Cease CL wear
  - Therapeutics
    - Start intensive topical antibiotics with loading dose
    - Monotherapy with fluoroquinolones or combination of fortified antibiotics (more severe)
    - Cycloplegics→ atropine or cyclopentolate
    - Oral analgesics
    - Never patch CL complications
  - Compliance: important→ needs intensive treatment and patient compliance
  - Culture→ eye, case, lens, solution
  - Review next day
    - Symptoms
    - Signs: epithelial defect? Infiltrate?
    - Worse? Revise DDx, microbes, resistance, compliance
  - Steroid? SCUT trial→ maybe later in the disease for more severe cases
  - Future contact lens wearer?

Acanthamoeba Keratitis

- Serious and difficult to treat
- PAIN even in early stages
- Typically inflammation around the corneal nerves→ Dendritic like or ring

Meibomian Gland Dysfunction

Clear lipid droplet	Blocked Gland Orifices	Large Block Gland	Lid Margin Frothing
			

Meibomian Gland Dysfunction (MGD)

- Symptoms
  - Mild transient dry eye symptoms
  - CL associated dryness, discomfort
  - CL intolerance that cannot be explained (e.g. by lens fit, deposits)
- Signs
  - Frank blockage of one or more MG orifices
  - Inability to express lipid from orifices
  - Cloud or cheesy secretion on gland expression
  - Associated signs
    - Frothing of TF at lid margin or canthus
    - Excessive fine debris in tear film
    - Unstable tear film short TBUT
    - Elevated, depressed or irregular MG orifices
    - Localised or general hyperemia at lid margin
    - Papillary hypertrophy (mild to mod)
    - Thickening of lid margins (chronic cases)
    - Excessive (oily) CL deposits
- Management
  - Treatment aimed at
    - Unblocking block MG orifices
    - Expressing abnormal lipid from glands
    - Improving flow of normal lipid
  - "lid hygiene", aim to raise ambient temperature transiently→ improved "flow" of abnormal lipid

1. Warm compresses
    - Soak a face cloth or similar in warm to hot water, then squeeze
    - Fold the cloth and hold it firmly over the eye until the temp cools
    - Repeat with other eye
2. Lid Scrubs
    - Half fill the basin with warm water
    - Add 3-4 drops of Johnson's Baby Shampoo no tears formula
    - Dip a cotton bud into the soapy water
    - Gently roll eyelid margin outwards and rub along it softly
    - Loosen and remove any scales of crusts from eyelid margins and bases of eyelashes
    - Repeat for upper lid and other eye
    - Products such as LidCare can be used
3. Gland Expression
    - Better done as in-office procedure
    - Usually reserved for severe cases
    - Firmly press on eyelid just below the lid margin: cotton bud inside may help
    - Observe the expressed MG secretion
    - Repeat
    - Aim to express as much of the contaminated secretion as possible
    - Repeat on later visits until MG resolves

- Course of treatment:
    - Warm compresses and lid scrubs performed by patient at home
    - 2x per day morning and evening
    - 2week initial period of treatment is recommended
    - recheck after 2weeks